

파이썬 14주차 출석과제

20231023 전도현

```
In [23]: import re # regex lib.
```

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In [24]: # search dials 1 (re.search())
p = re.compile(r'\d+\-\d+\-\d+') # "\-" == [-]
text = 'My number is 415-555-4242 or 010-1234-5678'
res=re.search(p,text)

print('Phone number found: '+res.group())
```

Phone number found: 415-555-4242

```
In [25]: # search dials 2 (re.findall() -> list)
p = re.compile(r'\d+\-\d+\-\d+') # "\-" == [-]
text = 'My number is 415-555-4242 or 010-1234-5678'
res=re.findall(p,text)

print('Phone number found: '+res[0])
```

Phone number found: 415-555-4242

```
In [26]: # string lookup
p = re.compile(r'Bat(man|mobile|copter)') # 'Bat' + contents in () will make
text = 'Batman uses a Batmobile rather than a Batcopter'
res = re.findall(p,text)

## add 'Bat' to every result since it's excluded
res = ['Bat' + match for match in res]
### if you don't want it: use r'Bat(?:man|mobile|copter)' as regex
print(res)
```

['Batman', 'Batmobile', 'Batcopter']

```
In [27]: # string lookup: complicated
# p = re.compile(r'Batman/Batwoman') # why?
p = re.compile(r'Bat(?:wo)?man') # flagging ?: at the very first inside () w
text = 'Batman loves Batwoman'
res = re.findall(p,text)
print(res)
```

['Batman', 'Batwoman']

```
In [28]: # string combined with intager and spaces
p = re.compile(r'\d+\s\w+') # some intagers, single spaces, some words in [a
text = '12 drummers, 11 pipers, 10 lords, 9 ladies, 8 maids, 7 swans, 6 gees
      5 rings, 4 birds, 3 hens, 2 doves, 1 paetridge'
res = re.findall(p,text)
print(res)
```

['12 drummers', '11 pipers', '10 lords', '9 ladies', '8 maids', '7 swans',
'6 geese', '5 rings', '4 birds', '3 hens', '2 doves', '1 paetridge']

```
In [29]: # AEIOU
p = re.compile(r'[aeiouAEIOU]')
text = 'RoboCop eats baby food. BABY FOOD.'
res = re.findall(p,text)
print(res)
```

```
['o', 'o', 'o', 'e', 'a', 'a', 'o', 'o', 'A', 'O', 'O']
```

```
In [30]: # then if not?
p = re.compile(r'^[aeiouAEIOU]')
text = 'RoboCop eats baby food. BABY FOOD.' # using same string
res = re.findall(p,text)
print(res)
```

```
['R', 'b', 'C', 'p', ' ', 't', 's', ' ', 'b', 'b', 'y', ' ', 'f', 'd', '.',
 ' ', 'B', 'B', 'Y', ' ', 'F', 'D', '.']
```

```
In [31]: # now exclude spaces too
p = re.compile(r'^[aeiouAEIOU\s]')
text = 'RoboCop eats baby food. BABY FOOD.'
res = re.findall(p,text)
print(res)
```

```
['R', 'b', 'C', 'p', 't', 's', 'b', 'b', 'y', 'f', 'd', '.', 'B', 'B', 'Y',
 'F', 'D', '.']
```

```
In [32]: # Wildcards
p = re.compile(r'.at')
text = 'The cat in the hat sat on the flat mat.'
res = re.findall(p,text)
print(res)
```

```
['cat', 'hat', 'sat', 'lat', 'mat']
```

```
In [33]: # match all (using .*)
p = re.compile(r'First Name: (.*?) Last Name: (.*?)') # things in () will match
text = 'First Name: AI Last Name: Sweigart'
res = re.findall(p,text)
print(res)
```

```
[('AI', 'Sweigart')]
```

```
In [34]: # ignore caps
p = re.compile(r'robocop\s\d+',re.I) # re.I (caps i) will ignore caps
text = 'Robocop 1, ROB0COP 2, r0b0c0P 3, roboCOP 4, RoBoCoP 5'
res = re.findall(p,text)
print(res)
```

```
['Robocop 1', 'ROB0COP 2', 'r0b0c0P 3', 'roboCOP 4', 'RoBoCoP 5']
```

```
In [35]: # * but not not greedy
p = re.compile(r'<.*?>')
text = '<To serve man> for dinner.>'
res = re.findall(p,text)
print(res)
```

```
['<To serve man>']
```

```
In [36]: # if greedy:
p = re.compile(r'<.*>')
text = '<To serve man> for dinner.>' # using text above
res = re.findall(p,text)
print(res)
```

```
['<To serve man> for dinner.>']
```

```
In [37]: # Replace some
p = re.compile(r'Agent \w+') # matches 'Agent ' and some words in a row
text = 'Agent Alice gave the secret documents to Agent Bob.'
res = re.sub(p, 'CENSORED', text) # compiled regex, replacing word, source
print(res)
```

```
CENSORED gave the secret documents to CENSORED.
```

```
In [38]: # dials
#####
# regex explain #
#####
# ((area code?)(sep?)(first)(sep)(last)(ext?)) #
#####
p=re.compile(r'''(
  (\d{2})? # 국번 (이때 모든 국번은 0으로 시작함)
  (\s|-|\.)? # sep
  \d{4} # 앞 4자리
  (\s|-|\.)? # sep
  \d{4} # 뒤 4자리
  (\s*(ext|x|ext.)\s*\d{2,5})? # extension
)''', re.VERBOSE)

text = '(082)-5900-7777, 010-1234-5678, 010 3333 8888'

res = re.findall(p,text)
print(list(map(lambda x: x[0],res)))
```

```
['(082)-5900-7777', '010-1234-5678', '010 3333 8888']
```

```
In [39]: # 재미로, 국제전화 함 해볼게요

p = re.compile(r'''(
  (\d{1,3})? # 국제전화 확장
  (\s|-|\.)? # sep
  |0 # 또는 0
  )\d{2}\)? # 국번 (국제전화 예외처리)
  (\s|-|\.)? # sep
  \d{4} # 앞 4자리
  (\s|-|\.)? # sep
  \d{4} # 뒤 4자리
  (\s*(ext|x|ext.)\s*\d{2,5})? # extension
)''', re.VERBOSE)

text = '''
+82-10-1254-8854
010 4853 4865
```

```
(010) 2258 7744  
(082)-5900-7777, 010-1234-5678, 010 3333 8888  
...
```

```
res = re.findall(p,text)  
print(list(map(lambda x: x[0],res)))
```

```
['+82-10-1254-8854', '010 4853 4865', '025 1487-4572', '(010) 2258 7744',  
'(082)-5900-7777', '010-1234-5678', '010 3333 8888']
```

In []: